

INTEGRATED ELECTRICAL SYSTEMS OF SOLAR PHOTOVOLTAIC TECHNOLOGY

LABORATORY 2

GROUP 2



TÉCNICO LISBOA

The Influence of Temperature in Photovoltaic Cells

Authors:

Francisco Teixeira ist1102662
Manuel Barreiras ist1102554
Beatriz Coelho ist1107113

franciscorfteixeira@tecnico.ulisboa.pt
manuel.barreiras@tecnico.ulisboa.pt
beatriz.h.coelho@tecnico.ulisboa.pt

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1 Introduction

The main goal of this laboratory is to understand how temperature influences the performance of photovoltaic cells. This effect will be analysed by applying different values of temperature to the solar cell using a Peltier module and a Lamp.

Using the acquired experimental data, I-V and P-V characteristic curves will be produced to compare the module's performance across the different temperatures. It is expected that an increase of temperature will lead to a reduction of the open-circuit voltage, V_{oc} . As the temperature increases, the semiconductor's bandgap decreases which allows a larger fraction of the incident photons to possess sufficient energy to generate electron-hole pairs, increasing slightly the short-circuit current. An increase in temperature elevates the rate of electron-hole recombination, therefore degrading the cell's capacity to maintain a high potential difference across its p-n junction, leading to a drop in the open-circuit voltage. The overall power output of the cell will decrease, as the drop in open-circuit voltage outweighs the slight increase in the short-circuit current. This behaviour can be visualised in Figure 1.

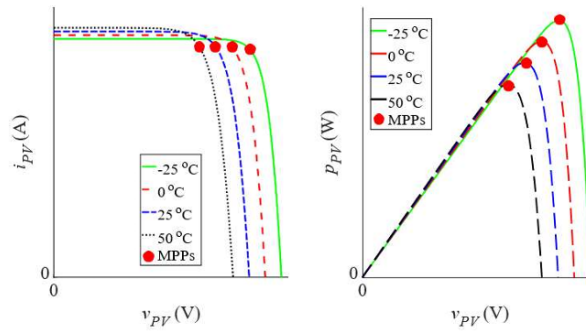


Figure 1: Expected experimental results

2 Results

All tests were conducted using a projector as the light source, placed at 30 cm from the table and the measured irradiance was 1020 W/m^2 , trying to match the irradiance from the previous lab (1015 W/m^2).

The setup consisted of the solar module placed on top of a Peltier module with the purpose of heating the solar module to test it at different temperatures. The module was connected in series to a variable resistor and a 1Ω resistor to measure current. The I(V) curve was captured performing a load sweep using the variable resistor. As the resistor was manually adjusted, the operating point of the solar cell transitioned through the entire curve from the short circuit point (lowest resistance) to the open circuit point (highest resistance).

In order to perform the I(V) curve, MATLAB was used to fit a smooth trend line into the points of the oscilloscope. A figure with Power vs Voltage was also performed in order to know the maximum power point of operation, where the output power is maximum.

2.1 DC Voltage versus Temperature of the Peltier Module

The first step of the experimental procedure was to characterize the Peltier module's temperature response to the applied DC voltage with and without the light projector. To ensure a great experimental characterisation, four distinct DC voltage values (0V, 1V, 2V and 3V), where each one corresponds

to a different value of temperature.

Table 1: Experimental results of Temperature as a function of DC Voltage

Voltage (V_{dc})	Without Lamp (T)	With Lamp (T)
0 V	18.5	50.4
1 V	24.0	54.7
2 V	30.5	59.3
3 V	39.0	63.9
Total Variation (ΔT)	20.5	13.5
Average Temperature	28.0	57.1

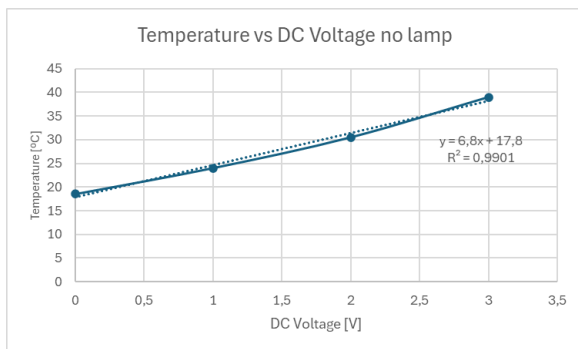


Figure 2: Evolution of temperature on the Peltier module with the light source turned off

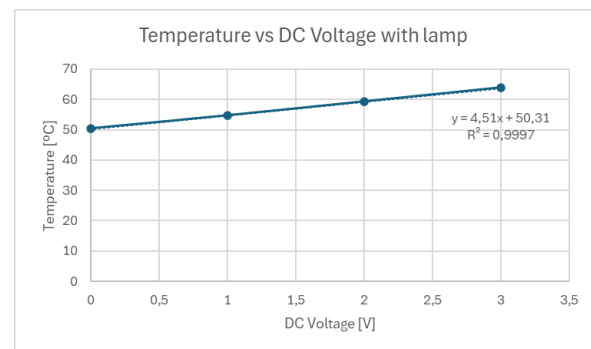


Figure 3: Evolution of temperature on the Peltier module with the light source turned on

As observed in Figures 2 and 3, the Peltier module presents a linear heating profile in response to the DC voltage applied. The presence of the light projector clearly increases the ambient temperature, reaching 50.4°C. Additionally, it is also noticeable that the slope of the curve decreases when the light projector is turned on. This reduction in the heating rate occurs because, at higher temperatures, the Peltier module's thermal losses to the environment increases.

The characteristic with the light projector is the appropriate choice for the experimental procedure. Since the light projector will be used during the procedure, the most reasonable option is to include it in the Peltier module's characterization to take into account the extra temperature that the lamp adds to the system.

2.2 I(V) and P(V) Photovoltaic Module Characteristics for different temperatures

Following the Peltier module's characterization, the cell module was tested under three specific temperature setpoints, chosen to match the previous temperature points from the characterization. The resulting I-V and P-V curves of each temperature were plotted.

Ambient temperature at 54.5°C

The results obtained when the ambient temperature was at 54.5°C were the following. Curves were obtained using MATLAB and the values of V_{oc} , I_{sc} and P_{max} are represented on the figures.

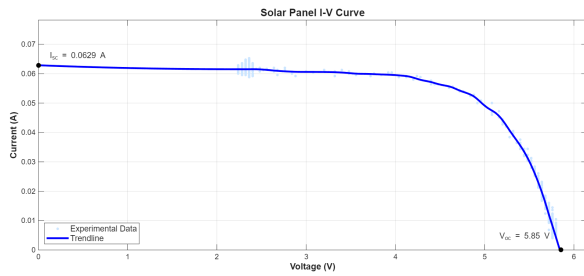


Figure 4: I-V curve for 54.5°C

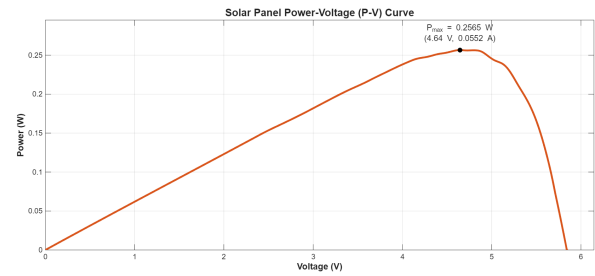


Figure 5: P-V curve for 54.5°C

Table 2: Electrical parameters at 54.5 °C

Parameter	Value
V_{oc} (V)	5.85
I_{sc} (A)	0.0629
P_{max} (W)	0.2565
V_{max} (V)	4.64
I_{max} (A)	0.0552

Ambient temperature at 60.3°C

The results obtained when the ambient temperature was at 60.3°C were the following. Curves were obtained using MATLAB and the values of V_{oc} , I_{sc} and P_{max} are represented on the figures.

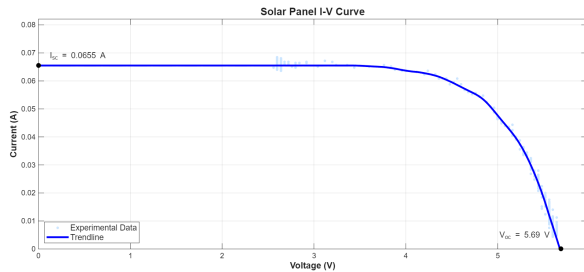


Figure 6: I-V curve for 60.3°C

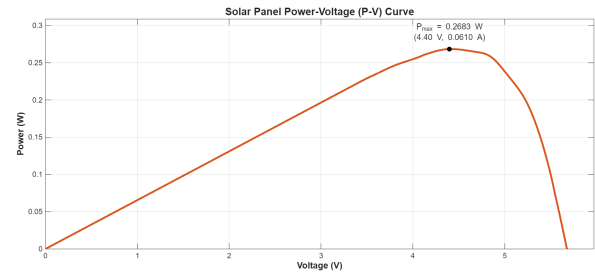


Figure 7: P-V curve for 60.3°C

Table 3: Electrical parameters at 60.3 °C

Parameter	Value
V_{oc} (V)	5.69
I_{sc} (A)	0.0655
P_{max} (W)	0.2683
V_{max} (V)	4.4
I_{max} (A)	0.0610

Ambient temperature at 63.9°C

The results obtained when the ambient temperature was at 63.9°C were the following. Curves were obtained using MATLAB and the values of V_{oc} , I_{sc} and P_{max} are represented on the figures.

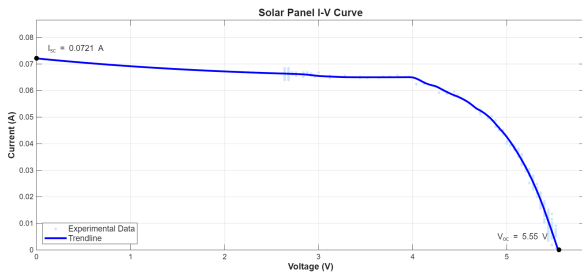


Figure 8: I-V curve for 63.9°C

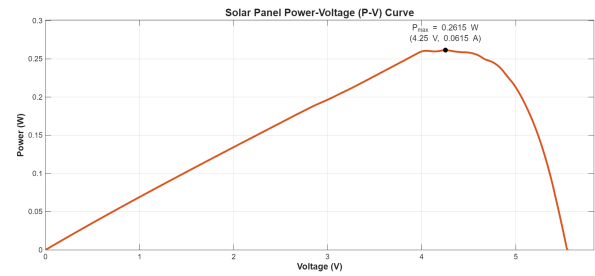


Figure 9: P-V curve for 63.9°C

Table 4: Electrical parameters at 63.9 °C

Parameter	Value
V_{oc} (V)	5.55
I_{sc} (A)	0.0721
P_{max} (W)	0.2615
V_{max} (V)	4.25
I_{max} (A)	0.0615

2.2.1 Combined I-V curves

To better visualize the effect of the temperature on the I-V curve and P-V, all curves were plotted on the same figure. The results obtained are presented below.

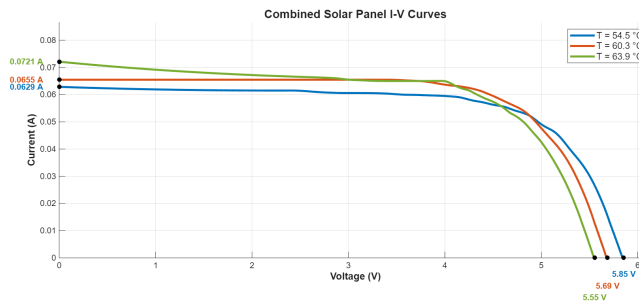


Figure 10: Combined I-V curves for different temperatures tested

Analysing the I-V curves for each value of temperature, the theoretical expectations are verified. It is possible to notice that the open-circuit voltage, V_{oc} , experiences a continuous reduction when the temperature is increased, which is explained by the increased recombination rates, reducing the cell's ability to maintain the same potential difference. Additionally, the short-circuit current reveals an increasing upon temperature. This behaviour also aligns with the theoretical prediction. With the decreasing of the bandgap, a larger amount of the incident photons have sufficient energy to generate electron-hole pairs.

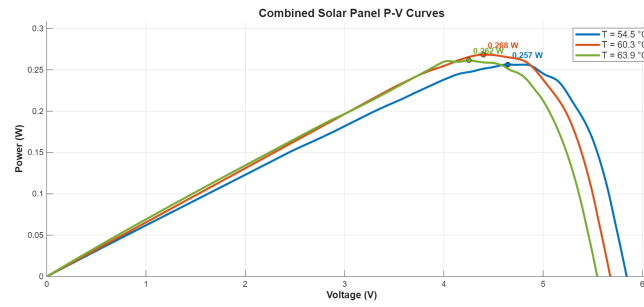


Figure 11: Combined P-V curves for different temperatures tested

Finally, the output power registered a slight increase instead of the expected decrease. This unexpected result can likely be attributed to experimental errors, such as fluctuations in the light projector's irradiance or a minor displacement of the solar cell during the measurements. These small variations could increase the incident light on the module, boosting the current values and attenuating the expected power losses. Also, the difference on temperatures tested was not very significantly, contributing to unexpected results. When temperatures are close to each other, the corresponding differences in maximum power are negligible.

2.3 Temperature Coefficients of Characteristic/Operation Points

For each parameter different curves were tested: linear, quadratic and exponential. After that the value of R^2 was compared and the curve with a highest R^2 was chosen. For linear fittings the temperature coefficient is the slope of the curve, but for non-linear curves a different approach was used. Because the average of the temperatures tested was 60°C , the temperature coefficient for non-linear curves was obtained from the slope of the tangent line to the curve at $T = 60^\circ\text{C}$.

Temperature Coefficient - V_{oc}

The open circuit's voltage for the temperatures tested were plotted in a figure. Then in MATLAB different curves were tested to see the one that would fit the most into the points registered. The results obtained were the following:

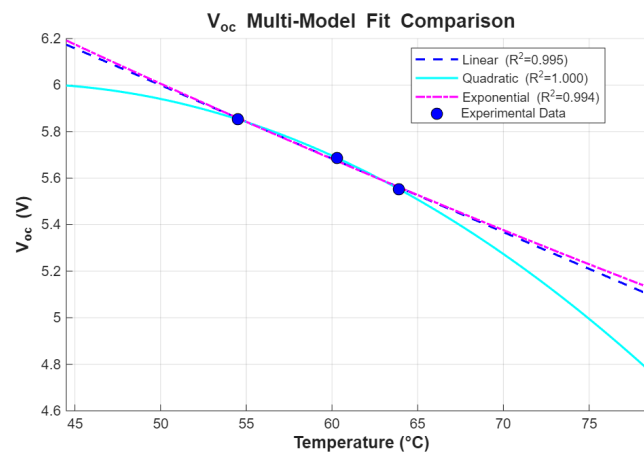


Figure 12: Multi-model fit comparison for V_{oc}

Table 5: Curve fitting results for the open-circuit voltage V_{oc} as a function of temperature.

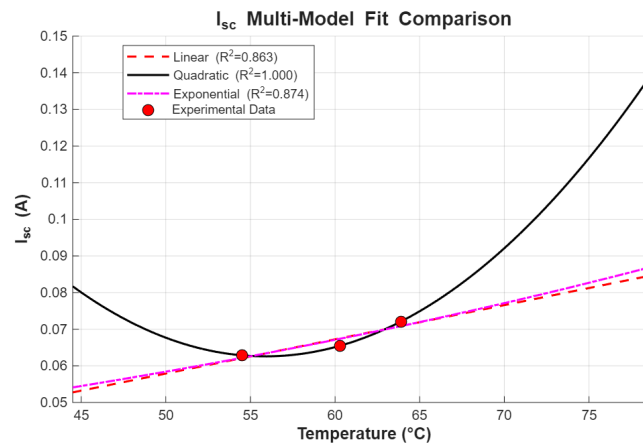
Model	R^2	Slope / Derivative ($V/^{\circ}C$)
Linear	0.99492	-0.0316
Quadratic	1.00000	-0.0333 (at $60^{\circ}C$)
Exponential	0.99388	-0.0315 (at $60^{\circ}C$)

From all the tested curves, the quadratic fit presented the highest R^2 value. Therefore, the quadratic model was selected. The temperature coefficient for V_{oc} obtained was:

$$\beta_{V_{oc}} = -33.3 \text{ mV}/^{\circ}C$$

Temperature Coefficient - I_{sc}

The short circuit's current for the temperatures tested were plotted in a figure. Then in MATLAB different curves were tested to see the one that would fit the most into the points registered. The results obtained were the following:

**Figure 13:** Multi-model fit comparison for I_{sc} **Table 6:** Curve fitting results for the short-circuit current I_{sc} as a function of temperature.

Model	R^2	Slope / Derivative ($A/^{\circ}C$)
Linear	0.86285	0.00093
Quadratic	1.00000	0.00122 (at $60^{\circ}C$)
Exponential	0.87435	0.00093 (at $60^{\circ}C$)

Although the quadratic curve had the highest R^2 , it's not expected the I_{sc} to increase quadratically with increasing temperature. The quadratic curve had a high value of R^2 because of the lack of data points, resulting in an overfitting problem. With just three points, a quadratic polynomial naturally forces a near-perfect fit, capturing experimental noise rather than the true underlying physical trend. The short-circuit current is expected to follow a more linear or exponential trend. Since the linear and exponential fits had similar slopes, the exponential model was chosen because it exhibited the highest R^2 .

$$\alpha_{I_{sc}} = 0.93 \text{ mA}/^{\circ}\text{C}$$

Temperature Coefficient - P_{max}

Maximum power for the temperatures tested were plotted in a figure. Then in MATLAB different curves were tested to see the one that would fit the most into the points registered. The results obtained were the following:

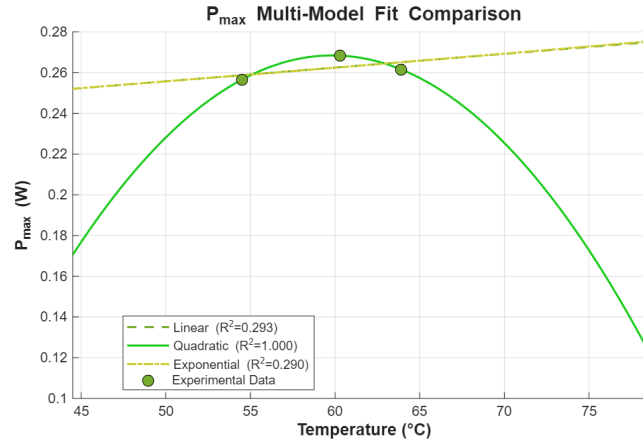


Figure 14: Multi-model fit comparison for P_{max}

Table 7: Curve fitting results for the maximum power P_{max} as a function of temperature.

Model	R^2	Slope / Derivative ($\text{W}/^{\circ}\text{C}$)
Linear	0.29257	0.0007
Quadratic	1.00000	-0.0001 (at 60°C)
Exponential	0.28951	0.0007 (at 60°C)

Because both V_{oc} and I_{sc} follow approximately linear trends, and power depends on both voltage and current, the resulting dependence of P_{max} on temperature tends to be quadratic. As explained before, the experimental results were not the expected, since power is theoretically expected to decrease with increasing temperature. The same problem of overfitting occurred here. Nevertheless, the quadratic model was still chosen for fitting, and it even yielded a negative temperature coefficient, consistent with the general decreasing trend of P_{max} .

$$\gamma_{P_{max}} = -0.1 \text{ mW}/^{\circ}\text{C}$$

2.4 Photovoltaic cell equivalent model

In order to model the electrical behaviour of the photovoltaic module for the three temperature conditions analysed previously, a single-diode five-parameter (1D5P) equivalent model was used. The experimental I(V) points acquired for each experiment were fitted in MATLAB in order to estimate the parameters of the model that best reproduce the measured curves. The quality of the fitting was evaluated through the Root Mean Square Error (RMSE).

The I(V) characteristic of the photovoltaic model can be described by

$$I = I_{ph} - I_{is} \left(e^{\frac{V - R_s I}{n u_T}} - 1 \right) - \frac{V + R_s I}{R_{sh}} \quad (1)$$

where I_{ph} is the photocurrent, I_{is} is the diode saturation current, R_s is the series resistance, R_{sh} is the shunt resistance, n is the ideality factor and u_T is the thermal voltage.

The extracted parameters for the three temperature conditions are presented in Table 8. The corresponding temperatures were obtained from the experimental characterisation presented in the previous section.

Table 8: Extracted 1D5P model parameters for the three temperature conditions.

Temperature (°C)	I_{ph} (A)	I_{is} (A)	R_s (Ω)	R_{sh} (Ω)	n	RMSE
54.5	0.06248	4.10×10^{-8}	3.09	3221.89	15.90	1.58×10^{-3}
60.3	0.06621	8.04×10^{-8}	3.23	31492.04	16.13	1.69×10^{-3}
63.9	0.06704	9.28×10^{-8}	2.97	4502.51	15.89	1.63×10^{-3}

It should be noted that the fitted ideality factor n corresponds to the complete photovoltaic module. Since the module is composed of several cells connected in series, the corresponding cell-level ideality factor can be estimated by dividing the fitted value of n by the number of series-connected cells, resulting in a value close to unity.

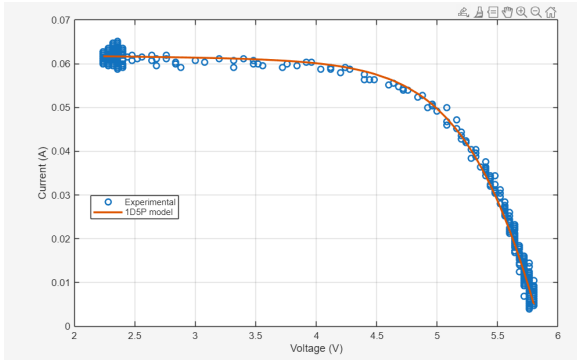


Figure 15: Experimental and modelled I(V) characteristic at 54.5°C.

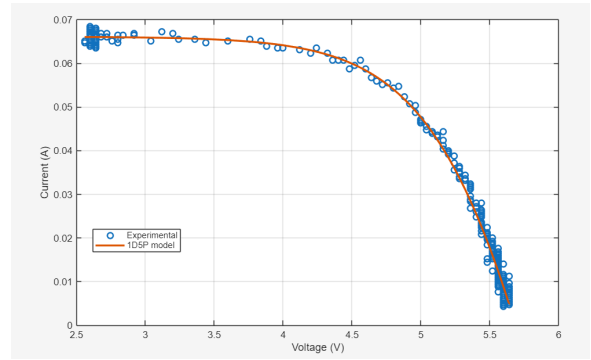


Figure 16: Experimental and modelled I(V) characteristic at 60.3°C.

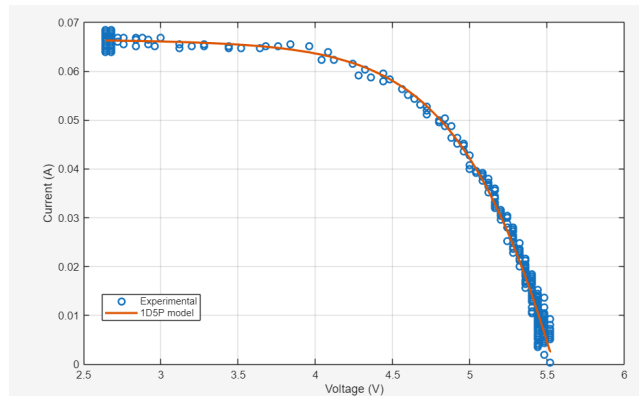


Figure 17: Experimental and modelled I(V) characteristic at 63.9°C.

From the fitted curves, it is possible to observe the expected thermal behaviour already discussed in the previous sections. As temperature increases, the open-circuit voltage decreases, which is visible in the progressive leftward shift of the end of the I(V) curve. At the same time, the short-circuit current increases slightly. This trend is consistent with the experimental values obtained previously, where V_{oc} decreases from 5.85 V to 5.69 V and then to 5.55 V as the temperature rises from 54.5°C to 60.3°C and 63.9°C, while I_{sc} increases from 0.0629 A to 0.0655 A and 0.0721 A.

The RMSE values are low for all three temperature conditions, ranging from 1.58×10^{-3} to 1.69×10^{-3} . This indicates a good agreement between the fitted model and the experimental data, and therefore confirms that the 1D5P equivalent model provides an adequate representation of the photovoltaic behaviour of the module under the analysed operating temperatures. Among the three cases, the lowest RMSE is obtained for 54.5°C, while the fittings for 60.3°C and 63.9°C remain very similar in quality.

2.5 Extraordinary Questions

2.5.1 LTSpice Simulation

To verify the experimental measurements and the theoretical model's output at different temperatures, a simulation of the single-diode five-parameter (1D5P) equivalent circuit was performed in LTSpice, as shown in Figure 18. Since the software can only compute one DC sweep at a time, all three circuits must share the same voltage source to plot their I-V curves on a single graph. All parameters implemented in the simulation were derived from the theoretical model.

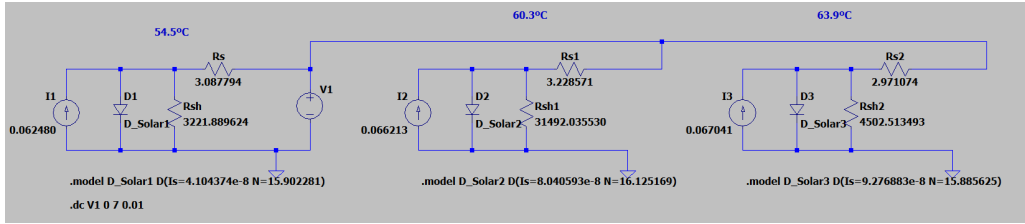


Figure 18: LTSpice equivalent circuit of model 1D5P for the different temperature values

Analyzing the simulated I-V curves in Figure 19 confirms the previously observed trends: with increasing temperature, the open-circuit voltage decreases linearly, while the short-circuit current increases exponentially. Furthermore, the simulated I_{sc} and V_{oc} values for the three temperature setpoints are in excellent agreement with both the experimental and theoretical results.

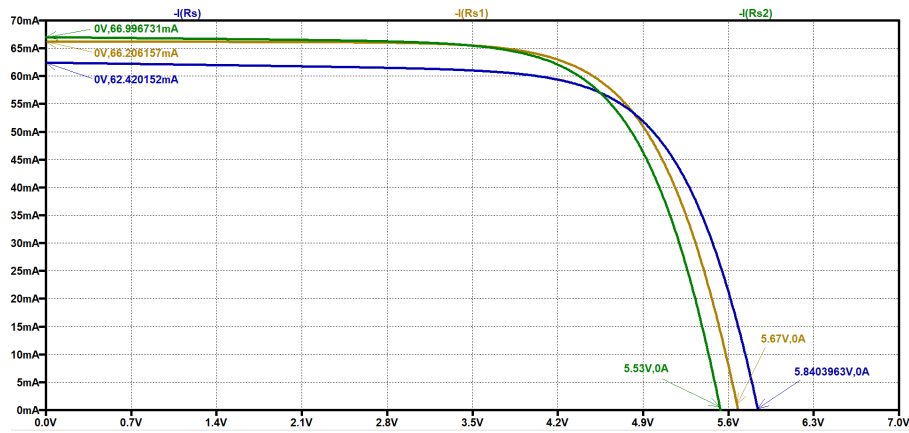


Figure 19: Simulated I-V curves for the three temperature values

Table 9: Comparison between experimental and simulated values of V_{oc} and I_{sc} across different operating temperatures.

Temp. (°C)	V_{OC} (V)			I_{SC} (A)		
	Exp.	Sim.	Error (%)	Exp.	Sim.	Error (%)
54.5	5.85	5.84	0.0017%	0.0629	0.0624	0.0080%
60.3	5.69	5.67	0.0035%	0.0655	0.0662	-0.0105%
63.9	5.55	5.53	0.0036%	0.0721	0.0670	0.0761%

The simulated output power also strongly aligns with the experimental data, Figure 20. However, as previously discussed, the theoretical maximum power should decrease as temperature rises, because the pronounced decrease in V_{oc} outweighs the slight increase in I_{sc} .

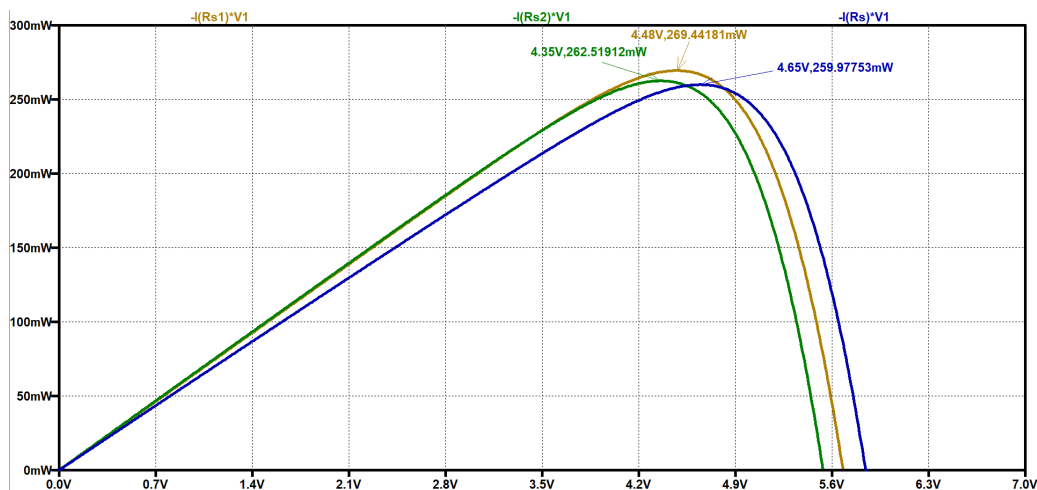


Figure 20: Simulated P-V curves for the three temperature values

Table 10: Comparison between experimental and simulated maximum power (P_{max}) across different operating temperatures.

Temp. (°C)	Exp. P_{max} (W)	Sim. P_{max} (W)	Error (%)
54.5	0.257	0.260	-0.0115%
60.3	0.268	0.269	-0.0037%
63.9	0.262	0.263	-0.0038%

3 Conclusion

This laboratory aimed to evaluate the influence of temperature on the performance of photovoltaic cells. By utilizing a Peltier module and a light source, the experiment characterized the I-V and P-V curves of a photovoltaic module under three temperatures: 54.5°C, 60.3°C, and 63.9°C.

Regarding the voltage and current, the experimental results verified the theoretical expectations. With an increase of temperature open circuit voltage should decrease and short circuit current should increase slightly.

Regarding maximum power, the experimental results did not correspond to the expectations. Power should decrease but the results obtained suggested a minor increase. This anomaly can be attributed to experimental errors, such as fluctuations in the projector's irradiance or minor displacements of the solar cell during measurements, combined with the fact that the small temperature variations tested yielded negligible power differences. In general, the results would have been more reliable if more data points had been collected over a wider range of temperatures. This would have helped to avoid overfitting in the curves and provided a clearer picture of the evolution of the parameters.

The implementation of the single-diode five-parameter model successfully replicated the experimental results with low RMSE values. The LTSpice implementation also replicated with accuracy the results with values close to both the experimental and theoretical results.